The Librarian

A cited question-answering service for Tunisian students going abroad

CS496 AI Engineering · Project 2

Oussama Sfar

Faten Toujani

Ines Turki

Feriel Charrad

Repository

https://github.com/SfarOussama9/the-librarian

Live service

https://sfaroussama9--librarian-web.us-east.modal.run

1. Overview

The Librarian answers questions from Tunisian students preparing to study in **France, Germany, Canada or an Erasmus+ programme**: deadlines, fees, required amounts, documents, scholarships and work rights. It answers **only from official documents**, cites every claim, and says **"I don't know"** when the documents don't answer.

281	**≈251k**	**2,200**	**125**
official documents, 11 publishers (Campus France, DAAD, IRCC, EU…)	words, French and English (266 pages, 15 PDFs)	chunks of ~150 words	eval questions: 110 answerable, 15 "I don't know"

2. Architecture

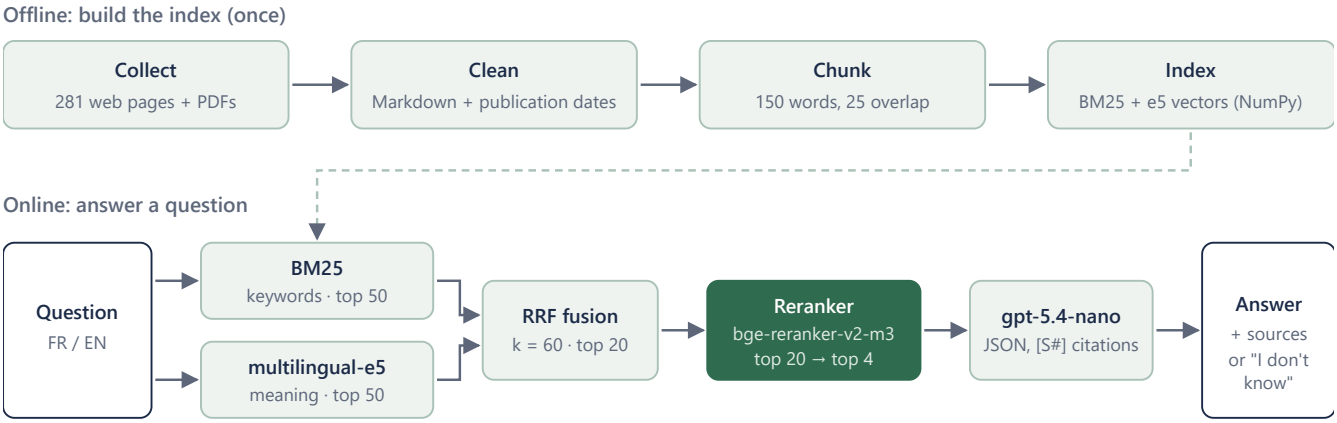


Figure 1. System architecture.

- **Chunking:** documents are split by heading, paragraph, list and table, so tables keep their header row. Each chunk keeps its URL, title, country, language, date and PDF page. We tested 150, 250 and 400 words; 150 found the right passage most often.
- **Index:** 2,200 vectors are searched exactly in under 10 ms from a NumPy file, so no vector database is needed at this size.
- **Generation:** the model answers only from the 4 chunks, cites `[S#]` after each fact, answers in the question's language, and prefers the newest dated source when two sources disagree.

3. Results

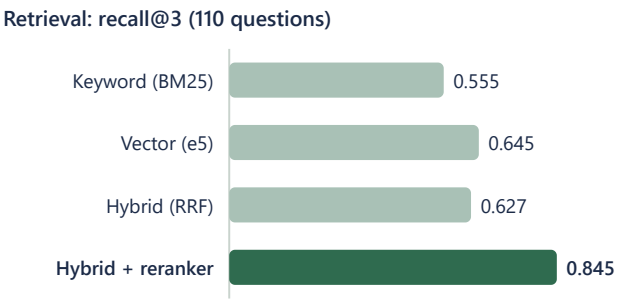


Figure 2. Ablation. The reranker adds +22 points over hybrid search.

Answers (125 questions, mean of 3 runs)	Final
Accuracy	**90.8%**
Faithfulness (claims supported by sources)	96.2%
Citation accuracy	94.6%
"I don't know" on unanswerable questions	97.8%

Graded by an LLM judge (gpt-5.4-mini) against reference answers and sources; we hand-checked 15 verdicts and agreed with 14.

- **The reranker is the largest gain.** Hybrid search usually finds the right chunk somewhere in its top 20; the cross-encoder, reading question and chunk together, moves it into the top 3.
- **Hybrid alone is not better than vectors alone,** but BM25 brings exact names and codes (KOSPIE, VLS-TS) into the candidates the reranker sees.

4. Week 8 optimisation

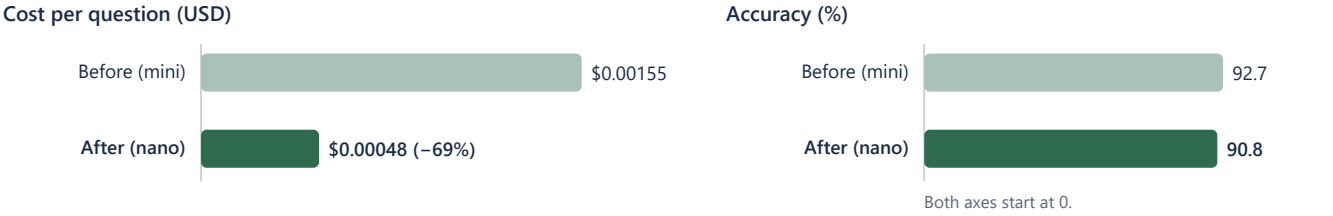


Figure 3. Target: −50% cost or p95 latency with at most 2 points of accuracy lost. Mean of 3 runs on the same 125 questions.

> **Cost −69% for −1.9 accuracy points.** We replaced gpt-5.4-mini with gpt-5.4-nano at high reasoning effort, and show dated sources newest first.

- **Tried and rejected:** nano at low effort (−5.9 points); worked examples in the prompt (−1.3 more); prompt caching (our prompt is under OpenAI's 1,024-token minimum, and nano didn't cache).
- **Latency could not be halved:** the answer step alone takes 3.4 s at p95, and faster rerankers lost 7–16 points of recall.

5. Limitations and cost

- **Cross-lingual misses:** a French question whose answer is only on an English page causes most wrong "I don't know" answers.
- **Outdated sources:** handled by dates, but 201 of 281 documents have no date, and nano still picks the old German blocked-account figure in French.
- **Online latency:** 12–22 s on Modal's free CPUs (reranker) vs ≈4 s on a laptop.
- **Cost model:** $0.48 per 1,000 questions, $48 at 100× (100,000). At that scale, hosting becomes the main cost.